Paleomagnetism of the Et-Then Group and Mackenzie Diabase in the Great Slave Lake Area[1]

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Studies of baked contact rocks and material at distances up to several kilometers from contacts show that the Mackenzie diabase acquired its remanent magnetization at the time it originally cooled about 1250 m.y. ago. Baked rocks have coercivities that are generally over twice as high as those of the diabase itself showing that they are suitable for paleomagnetic determinations. Combination of this new data (10 sites) with previous data from the Mackenzie Igneous Episode (Muskox Intrusion, Coppermine lavas, Mackenzie diabase) gives a mean pole position of 01 °N, 171 °W. Red beds of the Et-Then Group are shown to have acquired their magnetization prior to the intrusion of the diabase and probably at the time or soon after they were deposited. They have a mean direction of 294, −21 ($k = 17$, $\alpha_{95} = 10°$) and a pole at 01 °S, 048 °W ($K = 26$, $A_{95} = 8°$). The result is based on 14 sites (10 normal 4 reversed).

L'étude des contacts et du matériel prélevé jusqu'à plusieurs kilomètres des contacts montre que les diabases de Mackenzie ont acquis leur magnétisme remnant lors du refroidissement, il y a 1250 m.a. Les roches du contact ont une coercibileté généralement deux fois plus élevée que celle de la diabase, se prêtant ainsi aux déterminations du paléomagnétisme. En combinant ces nouvelles données (10 stations) avec les données antérieures de l'episode igné de MacKenzie, on obtient un pôle à 01° N et 171° W. Les couches rouges du Groupe Et-Then auraient acquis leur magnétisation avant l'intrusion des diabases, probablement lors du dépôt ou peu après. Elles ont une direction moyenne 294, −21 ($K = 17$, $\alpha_{95} = 10°$) et un pôle à 01° S, 048° W ($K = 26$, $A_{95} = 8°$). Ce résultat est basé sur 14 stations (10 normales et 4 inverses).

Introduction

Paleomagnetism will undoubtedly be of very great importance in determining the time order of Precambrian rocks, and in studying the nature of Precambrian orogenesis by determining any displacements that may have occurred between the provinces of the Canadian Shield. Before achieving these long term objectives, variations of the Precambrian geomagnetic field must be determined by conducting detailed studies across the entire shield. This work is a contribution to this general requirement.

Geology

The rocks studied occur in the east arm of the Great Slave Lake (Fig. 1). The region is part of the East Arm Fold Belt which is a subprovince of the Churchill Structural Province (Stockwell 1964). It is bounded on the south-east by the McDonald fault system and on the north by the Archean Slave Province. Within the Fold Belt a succession of folded sedimentary and volcanic rocks of the Great Slave Supergroup (Aphebian) overlies a basement of granitoid gneisses which yield Archean radiometric ages (Table 1). Dioritic and syenitic bodies, that yield three K-Ar biotite ages of 1630, 1795, and 1845 m.y. (Lowdon *et al.* 1963; Wanless *et al.* 1970), intrude the lower part of the Great Slave strata.

The Et-Then Group unconformably overlies the Great Slave Supergroup. It is a gently-dipping sequence of fluviatile sediments about 5000 m thick, and is divided into the Murky Formation beneath and the Preble Formation above. The Murky consists of massive conglomerates separated by thin lenses of red shale, siltstone, and sandstone. The Preble is composed for the most part of red or buff lithic and feldspathic sandstones. It attains a thickness of about 3000 m. Sedimentological studies of the conglomerates indicate that the transport direction was from south-east to north-west (Hoffman 1969). Apparently the McDonald fault system along the south-east margin was active at this time and produced

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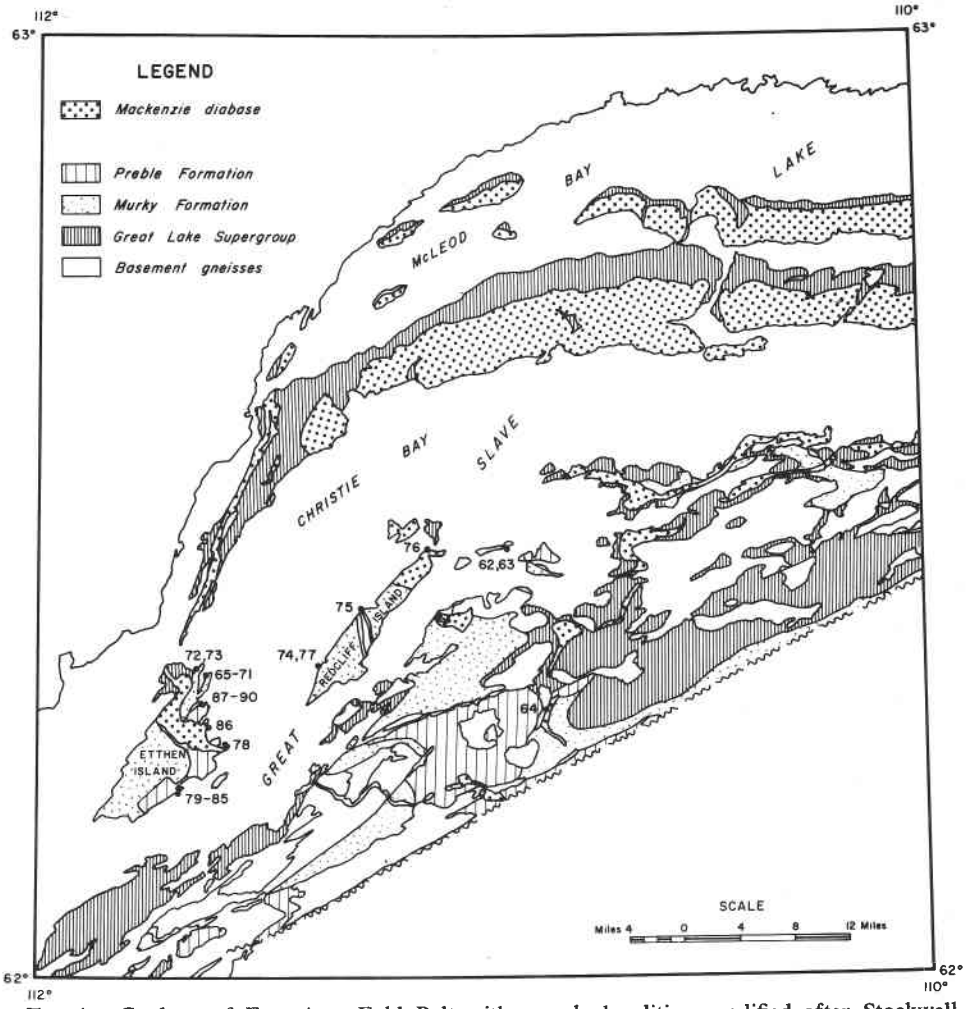


FIG. 1. Geology of East Arm Fold Belt with sample localities, modified after Stockwell (1936).

north-west facing scarps from which much of the Murky conglomerate was derived. Analysis of cross-bedding in the Preble sandstones indicates derivation from the north-east, as if it were deposited from rivers running parallel to the McDonald escarpment. The age of the Et-Then beds is not precisely known. They are younger than the diorites and syenites which intrude the Great Slave Supergroup, and older than the Mackenzie diabases (estimated mean age of 1250 m.y.) which cut them (Fahrig and Jones 1969). They have been tentatively correlated with somewhat similar sedimentary beds in the western Canadian Shield such as the Martin Formation, the Kazan Formation of the Dubawnt Group, and the Nonacho and Cameron Bay Groups (McGlynn 1970). Eventually it ought to be possible to determine the age relationships of the sediments in these supracratonic basins of the Western Shield from the paleomagnetic results.

The Et-Then and older rocks are intruded

TABLE 1. Stratigraphic section – East Arm Fold Belt

Mackenzie Diabase 1250 m.y.	
Et-Then Group	Preble Formation
	Murky Formation
Syenite, diorite 1630–1845 m.y.	
Great Slave Supergroup	
Archean basement	

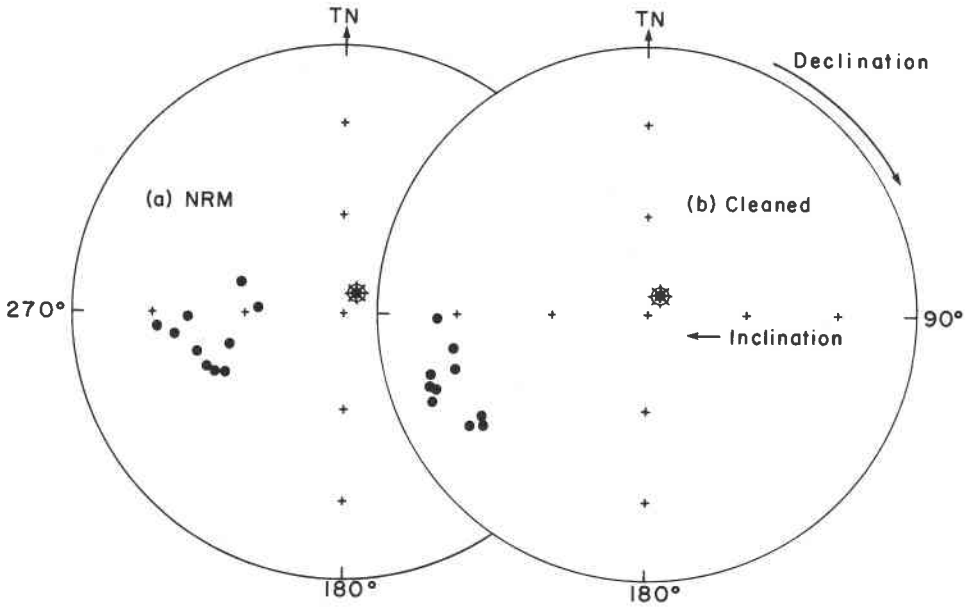


FIG. 2. Mean site directions of Mackenzie diabase. The primitive is the horizontal plane and the star is the present earth's field. The inclinations are all positive. (a) Initial n.r.m. directions. (b) Directions after magnetic cleaning.

by Mackenzie diabase. Sills, up to 100 m thick, occur in the large peninsulas in the northern part of the lake, and paleomagnetic results from 5 sites in these have been reported by Fahrig and Jones (1969). In the southern part of the basin, diabase occurs in moderately dipping sheets and small dikes and it is these that have been sampled in the present work.

Statement of Objectives

The first task in paleomagnetic work is to survey the directions of magnetization. The second is to determine when, during the history of the rock, the directions were acquired. Since the Et-Then sediments have been extensively intruded by diabase it is necessary to determine to what extent they have been remagnetized during the period of heating associated with these intrusions. It is useful also to determine the time at which the diabases themselves became magnetized. The answers to these questions can be obtained by making use of the classical igneous contact test (work of David reviewed in Bruhnes 1906, Irving 1964, Everitt and Clegg 1962). Therefore a study of the diabase, of unaltered Et-Then sediments, and of the Et-Then sediments baked by the diabase has been made.

Methods

Generally, about 5 oriented cores, 2.5 cm in diameter, have been obtained from each site. Where the rock was hard usually 2 or 3 hand samples have been collected. The site numbers (Fig. 1) are arbitrary and have no stratigraphic significance. Measurements of intensity and direction of magnetization have been made using a commercial spinner magnetometer. Two specimens have been prepared and measured from each core or hand sample. In the analysis, averages for each core or hand sample have been used as the basic statistical unit. The directions have been analyzed by the method of Fisher (1953). Geometrical averages of the intensities have been obtained using the method of Irving *et al.* (1966). Both alternating field (peak field in oersteds are quoted) and steady field demagnetization have been used to obtain the coercive force spectrum. For studies of the blocking temperatures, thermal demagnetization has been used. The apparatus has been described by Roy *et al.* (1969, 1971, 1972 (in

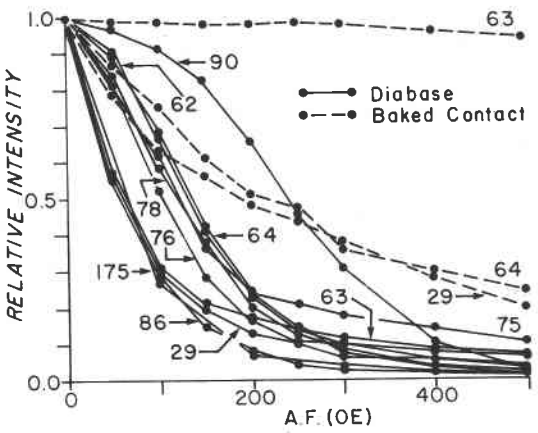


FIG. 3. Relative intensity versus alternating field (A.F.) for diabase and contact rocks.

preparation)). Demagnetization curves have been obtained from at least one specimen per site. In order to remove components of low stability, partial demagnetization (cleaning) has been carried out on all specimens, the demagnetizing conditions being selected using standard procedures (As and Zijderveld 1958: Irving *et al.* 1961). Measurements of the coercivity of remanence have been made usually on one specimen from each site. The coercivity of remanence is defined as the median coercive force of the particles present, and is the strength of the demagnetizing field required to reduce the magnitude of remanence by one half. For igneous rocks the coercivity of saturation isothermal remanent magnetization (i.r.m.s.), natural remanent magnetization (n.r.m.), and anhysteretic remanent magnetization (a.r.m.) have been determined. The a.r.m. was imposed in an alternating field (A.F.) of 2900 oe, and a bias field of 0.3 oe. For sediments, only the coercivity of i.r.m. has been measured, since the coercivities of n.r.m. and a.r.m. for material of such high coercivity cannot generally be measured with our apparatus. In the studies of i.r.m. a field of 10 000 oe is used, which is adequate to saturate the igneous rocks, but not the red sediments.

Mackenzie Diabase

Directions of magnetization from the sills (5 sites Fig. 10) north of Christie Bay have been reported previously by Fahrig and Jones (1969). We have studied 8 sites in the area of Et-Then and Redcliffe Islands to the south of Christie Bay (Fig. 1), and also two dikes—one (site 175) near Great Bear Lake, and one (site 29) west of Baker Lake (Fig. 10).

The mean directions of n.r.m. at 10 sites are plotted in Fig. 2(a). A.F. demagnetization curves (Fig. 3) show the regular decay that is characteristic of magnetically stable diabase (Stott 1963). During demagnetization, the directions at first move away from the direction of the present earth's field, and then remain stable in the range 200 to 500 oe. An example of magnetic cleaning at one site is given in Fig. 4(a). The mean standard deviation at the 10 sites is initially 19° and is reduced to 6° after cleaning. Cleaning also improves the agreement between sites, and causes the site mean directions to move systematically away from the present field (compare Fig. 2(a) and (b)). The site directions after cleaning are listed in Table 2.

At locality 29 (Fig. 10) cores have been obtained from the margin to the center of a thick dike. The average coercivity of a.r.m. at the fine-grained contact is low (80 oe), increasing to 110 oe at 1 m, and 130 oe at 3 m from the margin. Thereafter toward the center the coercivity remains at an average of about 130

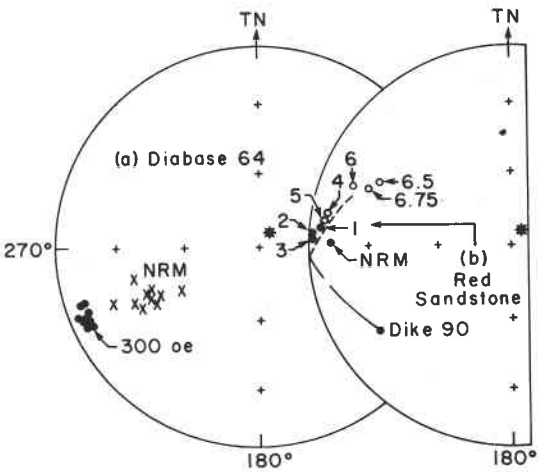


FIG. 4. Examples of (a) magnetic cleaning of diabase and of (b) thermal cleaning of red sandstone 4 m from dike 90. The primitive is the horizontal, and the direction of the present field is marked by a star. In (b) the numbers refer to demagnetization steps in 100 °C. The great circle between the direction of the dike and the stable direction in the sandstone is marked.

TABLE 2. Summary of magnetic properties of diabase and contact rocks

Site		n	M	h	D, I	Pole	coercivities		
							n.r.m.	a.r.m.	i.r.m.
62	diabase sill (62.5, 110.9)	4	3340	300	250, + 17	01S, 179W	116	136	252
63	diabase sill	5	1600	300	241, + 22	—	064	073	175
63	fine-grained baked sandstone	4	2290	2900	227, + 34	—	1200	—	1785
63	combined (62.5, 110.9)	9	—	—	235, + 28	02S, 164W	—	—	—
64	diabase sill	5	4520	300	248, + 09	—	118	121	224
64	baked conglomerate	3	19	400	252, + 30	—	206	271	512
64	combined (62.3, 110.8)	8	—	—	250, + 17	02S, 179W	—	—	—
75	diabase lopolith (62.4, 111.3)	4	1860	500	237, + 28	01S, 165W	110	144	257
76	diabase sill (62.5, 111.1)	5	3740	300	253, + 18	0, 181W	102	151	273
78	diabase sill (62.2, 111.6)	5	1950	300	247, + 15	04S, 177W	110	117	228
86	diabase sill (62.3 111.6)	5	3170	200	259, + 28	08N, 185W	068	087	173
90	diabase dike 0.5m (62.3, 111.6)	4	6330	300	237, + 23	04S, 167W	240	266	439
29	diabase dike 60m	9	1670	300	268, + 22	—	061	123	250
29	baked gneiss	8	269	300	268, + 24	—	194	255	539
29	combined (64.1, 97.1)	17	—	—	268, + 23	10N, 180W	—	—	—
175	diabase dike (66.9, 119.1)	3	1990	300	253, + 27	07N, 188W	068	148	276

The locality, latitude (N), and longitude (W) are given in the second column. n is the number of cores or hand samples (2 specimens measured from each), M is the intensity of n.r.m. $\times 10^{-6}$ cgs/cm^3, h is the cleaning field in oersteds, D, I are the declination and inclination of the mean site direction after cleaning. The coercivities (the median destructive field (Park and Irving 1970)) of n.r.m., a.r.m. (2900 oe, A.F. bias field 0.3 oe), and i.r.m. (10 000 oe) are given in oersteds.

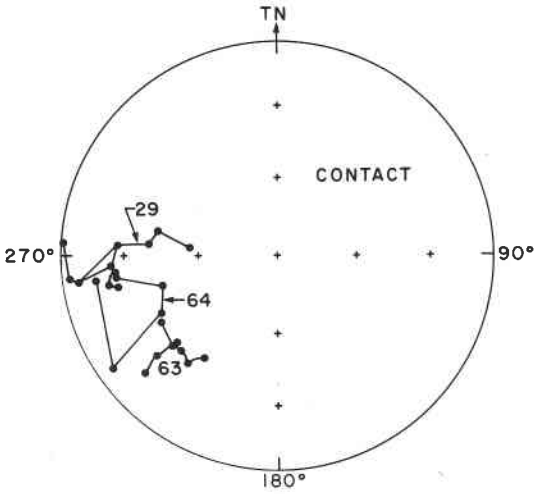


FIG. 5. Directions of magnetization in contact rocks at 3 sites after magnetic cleaning. The directions of all individual specimens are plotted. The primitive is the present horizontal.

oe. The highest coercivity is found at dike 90 (Table 2), which is thin and very fine-grained. Steady field demagnetization of diabase from each site is plotted in Fig. 6. The mean coercivity of n.r.m. for all diabases (individual values listed in Table 2) is 106 oe, of a.r.m. 137 oe, and of i.r.m.(s) 250 oe. The mean i.r.m.(s) is 0.89 cgs/cm^3.

Baked Contact Rocks

Samples close to 4 diabase bodies have been collected. Their directions are given in Fig. 5. At site 63, four samples of very fine-grained metamorphosed sediment were collected from immediately above a thick sill about 30 m thick. Their magnetization was extremely stable. Their coercivities exceed 1000 oe (Table 2) and the directions of n.r.m. are not changed within the experimental error after demagnetization in A.F. of 2900 oe. At site 64, three samples of silicified Murky conglomerate consisting largely of gneiss pebbles were collected from an inclusion near the upper contact of the diabase sill, and at site 29, eight samples of red granite (Hudsonian) ranging from 0 to 3 m from the contact were collected. The coercivities of these baked crystalline rocks, although lower than that in the baked fine-grained sediment (Fig. 6), are still greater than in the adjacent diabase. It is clear as has been found by other workers (e.g. Leng 1955) that baked contacts are very desirable material paleomagnetically.

At site 29 the directions in diabase and contact rocks agree perfectly (Table 2). At sites 63 and 64 there are substantial differences although the general trend is the same. Such differences between contact rocks and diabase have been reported by Graham and Hales (1957) in the Karroo Dolerites and were attributed by them to disturbance of the local field by the diabase during cooling of the contact rocks. A variety of interactions may occur between the field of material (contact or diabase) as it acquires a remanent magnetization and the material that is last to become magnetized. Little is known about these effects.

Red sediment from the Preble Formation has been sampled at a distance of 4 m from a fourth body, a thin (0.5 m) dike (90). The sediment is not visibly altered. The initial directions (Fig. 4) are to the west, about 30° from those in the diabase. During thermal demagnetization up to 500 °C the directions change as if a component due to the present field is

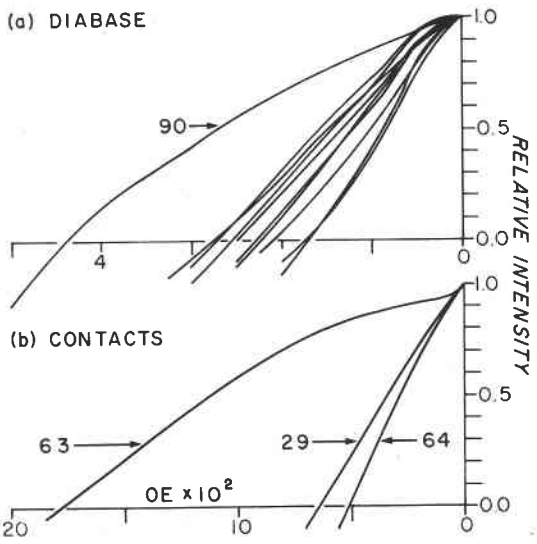


FIG. 6. Steady field demagnetization of (a) diabase and (b) contact rocks showing the high stability in contact rocks. In (a) 90 is the thin dike which is the most stable diabase. In (b) 63 is baked sediment, 29 baked gneiss, and 64 baked conglomerate.

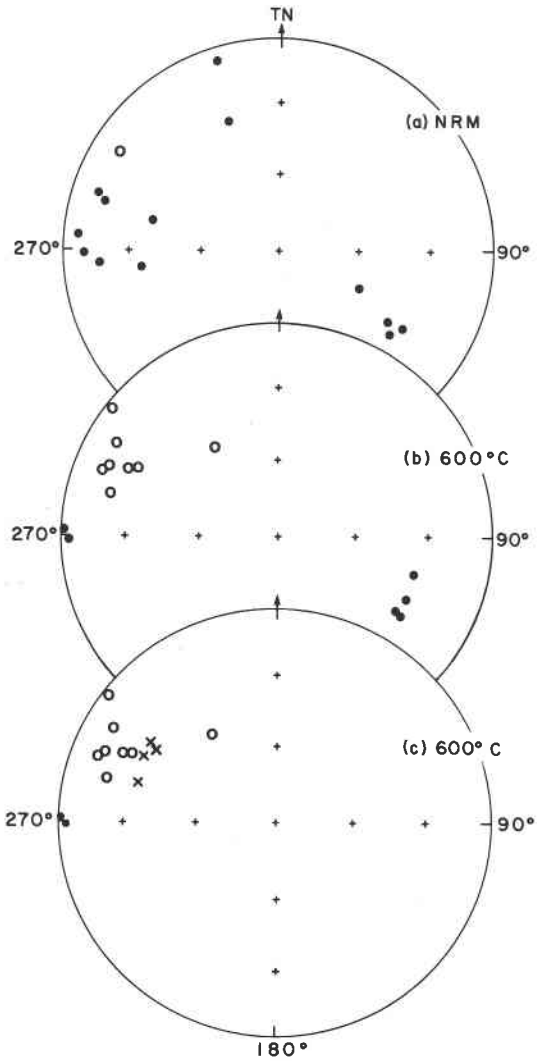


FIG. 7. Mean site directions of magnetization in the Et-Then Group. The primitive is the bedding plane. In (a) and (b) positive (negative) directions are marked as solid (open) dots. In (c) the directions are plotted irrespective of sign.

being removed. In the range 500–600 °C the directions move along a great circle away from the stable direction of the dike, that is, a component along the Mackenzie diabase direction is being removed. This is probably a partial thermo-remanent magnetization acquired during cooling after heating by the diabase. In the range 600 to 675 °C there is a stable end-point (WNW with negative inclination) which is parallel to the stable direction observed in the Et-Then sediments sampled at greater distances from the diabase, as is explained below.

Et-Then Group

Samples of fine-grained red sandstones were selected. The initial magnetization directions are generally widely scattered (Fig. 7). At 6 sites (79 to 81 and 83 to 85) from the Preble Formation, the specimens have directions that are grouped approximately around the present field (Fig. 8). Their intensities fall rapidly when thermally demagnetized (Fig. 9b), and the directions become random in the range 300 to 500 °C. At the other 14 sites the initial directions depart significantly from the present field (Fig. 7(a)), the thermal decay curves fall less rapidly, and reversed samples often show an increase in intensity due to removal of the component due to the present field (Fig. 9). After cleaning at 600 °C the directions become grouped around a WNW negative ESE positive axis. Ten sites are reversely and 4 are normally magnetized (Table 3, Fig. 7(b)). Because the variations in the geological dips are low the precisions before and after correction for dip are not significantly different (Table 4). The Preble is generally less stable than the Murky Formation, their mean coercivities of i.r.m.(s) being 5150 and 4280 oe respectively. The magnetization directions are summarized

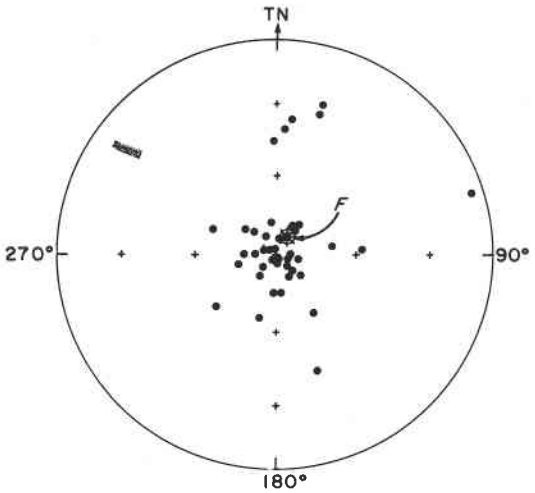


FIG. 8. The n.r.m. direction in Preble Formation unstable samples (sites 79 to 81 and 83 to 85). Specimen directions are plotted.

TABLE 3. Results from Et-Then Group

Site	n	M	D, I	Pole
PREBLE FORMATION (62.3N, 111.6W)				
82	3	6.3	325, − 47	05N, 081W
87	5	4.6	293, − 15	04S, 045W
88	5	5.3	295, − 24	0, 049W
89	4	5.2	285, − 20	02N, 040W
90	1	8.0	298, − 28	01N, 053W
MURKY FORMATION (62.3N, 111.6W)				
65	3	1.6	271, 0	0, 023W
66	2	2.7	269, + 03	0, 020W
67	2	1.7	308, − 01	16S, 056W
69	2	4.9	301, − 13	08S, 052W
70	2	1.4	291, − 13	04S, 043W
72	2	8.0	116, + 33	04N, 052W
73	2	6.4	106, + 34	09N, 045W
74	2	8.7	122, + 34	03N, 058W
77	2	12.1	123, + 32	01N, 058W

n is the number of samples (2 specimens from each), M is the intensity of n.r.m. $\times 10^{-6}$ cgs/cm^3, D, I declination and inclination of the mean direction after thermal cleaning at 600 °C.

in Table 4 and Fig. 7(c). The directions of stable magnetization are approximately constant throughout the group. There is no relationship between the directions of magnetization and the directions of flow of the water in which they were laid down (see above).

Age of Magnetization

Samples taken a few meters or less away from diabase are all magnetized in the same direction as the diabase. The intensity of magnetization of baked fine-grained Et-Then sandstone is several hundred times greater than in unbaked sandstone far from diabase (compare the intensity at site 63 in Table 2 with those in Table 3). The directions in sediments at distances greater than several hundred meters are different from those in the diabase. Sandstone collected at a distance of 4 m from diabase dike 90 which has been warmed (but not visibly altered) by intrusion of the diabase contains a component directed along the diabase direction. This component can be removed by heating to 600 °C, above which it reverts to the direction characteristic of sediment at a greater distance. Thus the stable magnetization of this diabase (that left after AF cleaning) was acquired at the time it cooled approximately 1250 m.y. ago, and the stable component of Et-Then sediments (magnetization present after thermal cleaning at 600 °C) was acquired prior to the intrusion of the diabase. There are no known heating events

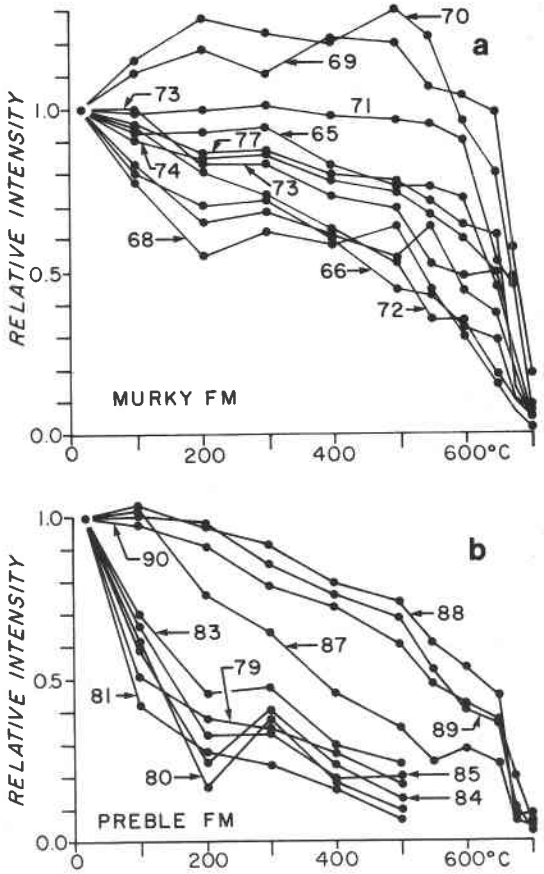


FIG. 9. Thermal demagnetization curves for Et-Then Group (a) Murky and (b) Preble Formations. Intensities are normalized to unity at room temperatures.

between the deposition of the Et-Then and the intrusion of the diabase, and this, and the presence of reversals shows that systematic components of the type characteristic of secondary magnetization are absent. Hence the magnetization of the sediments was probably acquired at the time of deposition and lithification, although this is not a firm conclusion.

Magnetic Field during the Mackenzie Igneous Episode

In the last decade there has been a steady accumulation of evidence suggesting that the Coppermine Lavas, the Muskox Intrusion, and the Mackenzie dike swarms are the extrusive, plutonic, and hypabyssal members of one huge

TABLE 4. Summary of Et-Then directions

		N		k	α_{95}
1.	Preble Form directions	5	298, − 27	22	17
2.	Preble Form poles	5	01N, 053W	25	16
3.	Murky Form directions	9	291, − 18	15	14
4.	Murky Form poles	9	01S, 045W	25	11
5.	Et-Then Group directions	14	**294, − 21**	**17**	**10**
6.	Et-Then Group directions	14	294, − 31	22	9
7.	Et-Then Group poles	14	**01S, 048W**	**26**	**8**

N is the number of sites, k is the precision, α_{95} is the radius of the circle of confidence in degrees ($P = 0.05$). The directions are defined by declination and inclination and the poles by latitude and longitude. All directions are corrected for bedding except entry 6 where they are calculated with respect to present horizontal. The values in bold face are regarded as the best summary of the data.

igneous province extending from the Arctic coast perhaps as far as Sudbury. The paleomagnetic results are summarized in Table 5 and Fig. 10. The first results (entry 1 of Table 5) were obtained from one dike (8 samples) of the Sudbury swarm. Later studies (entry 2, 13 samples from an unspecified number of dikes, and entry 3, 53 cores from 11 dikes) showed that the entire Sudbury swarm has essentially the same direction with very good agreement among dikes. Next, 21 dikes (entry 4 about 45 samples) from in and near the Muskox Intrusion were sampled, and two studies (entry 5, 22 sites of about 37 samples, and entry 6, 14 sites of 54 samples) extended the sampling to the south and east. All these studies of the Mackenzie diabase yield essentially the same direction, but it is not possible to combine the results in a uniform way because the sample coverage is variable (1–5 samples per site) and the site directions are not listed in some originals. Two combinations are given in Table 5 representing the limiting cases. In the first (entry 11) all the published site mean directions are combined giving each unit weight ($N = 35$). In the second (entry 12) the average of the five major studies are combined ($N = 5$). The mean poles differ hardly at all, but the areas of the error circles differ by a factor of 7. The error in entry 12 is a conservative estimate, whereas that in entry 11 is probably too small.

Also given in Table 5 are the results from the Muskox Intrusion (65 sites, 14 rock groups the latter given unit weight), the Coppermine Lavas (21 sites), and a combined result (unit weight to plutonic, extrusive, and hypabyssal elements ($N = 3$)). The data are not yet on a sufficiently uniform basis for detailed statistical testing, but it is clear that the mean direction of magnetizations of each is essentially identical.

The A.F. studies of Larochelle (1966) indicated that the diabase was stable magnetically, and Sopher (1963) and Robertson (1969) showed that some samples of baked country rock were magnetized approximately parallel to the diabase. Parallel magnetizations in igneous body and baked country rock could also be produced by later regional remagnetization, and so do not by themselves indicate a primary magnetization (Everitt and Clegg 1962). One of the objectives in the present work was to study this point and it has been shown that at least in the diabase of the East Arm of Great Slave Lake the magnetization is primary. Hence it now appears certain, because of (1) the excellent agreement over such a wide area, (2) the contact test, and (3) the high stability of the diabase and particularly of the baked country rock, that the magnetization of the intrusive, volcanic, and hypabyssal elements of this igneous episode were acquired at much the same time during cooling. The results are not yet sufficiently detailed to determine if small differences in directions occur between the elements. Hence there is no reason from the paleomagnetic evidence to believe that they differ in time.

The directions in the Et-Then are different from any yet observed on the Canadian Shield. Their significance will not become clear until very much more data from Proterozoic rocks is available.

Acknowledgments

We wish to acknowledge much help from Guy Gagnon in making the collections. We are grateful also to Dr. J. A. Donaldson for collecting site 175 and for critically reading the manuscript.

TABLE 5. Summary of results from Mackenzie diabase, Muskox intrusion, and Coppermine lavas

		N	Pole	K	Error	Reference
1.	Sudbury dike (266, − 08)	1	—	—	—	Strangway 1961
2.	" dikes (253, − 01)	13	08S, 163W	—	(4, 7)	Sopher 1963
3.	" " (268, − 01)	11	02S, 171W	210	3	Larochelle 1966
4.	Mackenzie dikes at Muskox (230, + 25; $\alpha_{95} = 14°$)	21	01N, 160W	—	(8, 15)	Robertson 1969
5.	" dike localities (2 of fig. 10)	22	08N, 177W	8	10	Fahrig *et al.* 1965
6.	" diabase " (1 " " 11)	14	04N, 167W	82	4	Fahrig and Jones 1969
7.	" " sites (10)	10	01N, 176W	68	6	present work
8.	" " Great Slave Lake (5 sites)	5	0, 168W	128	7	Fahrig and Jones 1969
9.	" " " " " (8 sites)	8	01S, 175W	83	6	present work
10.	results 8 and 9 combined	13	0, 172W	89	4	—
11.	Mackenzie diabase all available sites (35)	35	01N, 171W	78	3	—
12.	results 3, 4, 5, 6, 7 combined	5	02N, 170W	105	8	
13.	Muskox Instrusion	14	09N, 170W	25	8	Robertson 1969
14.	Coppermine Lavas (243, + 22; $\alpha_{95} = 5°$)	21	01N, 176W	—	(3, 5)	Robertson 1969
15.	results 12, 13, 14 combined	7	03N, 171W	122	6	—

N is the number of units used in statistical analysis. K is the precision of poles. The error is either the radius of the circle of confidence ($P = 0.05$) or dm, dp the semi-axes of the oval of error. These correspond to calculations based on models B and A respectively (Irving 1964, p. 49).

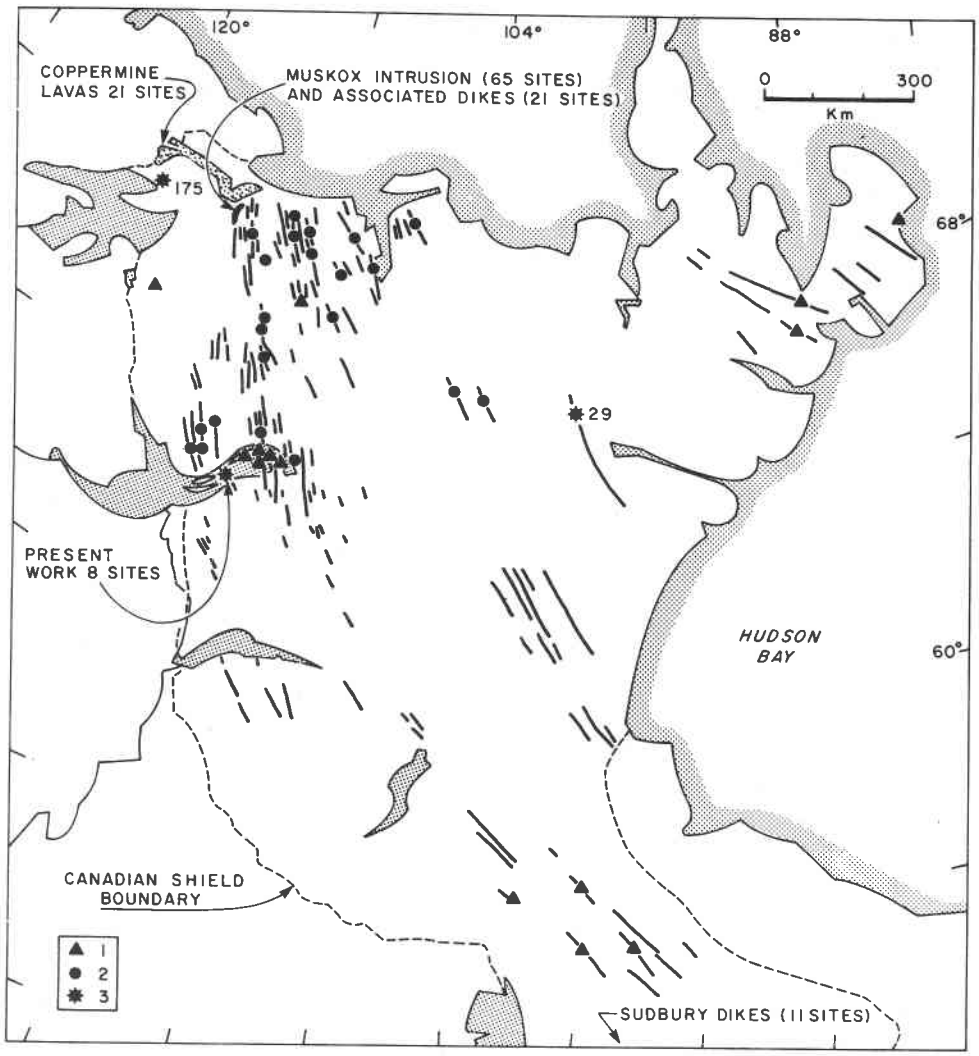


FIG. 10. Sketch map of the western Canadian Shield showing the locations of paleomagnetic sampling sites of the Mackenzie dike swarm. (1) Fahrig and Jones (1969), (2) Fahrig *et al.* (1965), (3) Present work.

AS, J. A. and ZIJDERVELD, J. D. A. 1958. Magnetic cleaning of rocks in palaeomagnetic research. Geophys. J., **1**, pp. 308–319.

BRUHNES, B. 1906. Recherches sur le direction d'aimantation des roches volcaniques. J. Phys., **5**, pp. 705–724.

EVERITT, C. W. F. and CLEGG, J. A. 1962. A field test of palaeomagnetic stability. Geophys. J., **6**, pp. 312–319.

FAHRIG, W. F., GAUCHIER, E. H., and LAROCHELLE, A. 1965. Paleomagnetism of diabase dykes of the Canadian Shield. Can. J. Earth Sci., **2**, pp. 278–298.

FAHRIG, W. F. and JONES, D. L. 1969. Paleomagnetic evidence for the extent of Mackenzie igneous events. Can. J. Earth Sci., **6**, pp. 679–688.

FISHER, R. A. 1953. Dispersion on a sphere. Proc. Roy. Soc. London, **217**, pp. 295–305.

GRAHAM, K. W. T. and HALES, A. L. 1957. Palaeomagnetic measurements on Karroo dolerite. Phil. Mag. Supp. Adv. Phys., **6**, pp. 149–161.

HOFFMAN, P. F. 1969. Proterozoic paleocurrent and depositional history of East Arm Fold Belt, NWT. Can. J. Earth Sci., **6**, pp. 441–462.

IRVING, E. 1964. Paleomagnetism. Wiley, 399 p.

IRVING, E., MOLYNEAUX, L., and RUNCORN, S. K.

1966. The analysis of remanent intensities and susceptibilities in rocks. Geophys. J., **10**, pp. 451–464.

IRVING, E., STOTT, P. M., and WARD, M. A. 1961. Demagnetization of igneous rocks by alternating fields. Phil. Mag., **6**, pp. 225–241.

LAROCHELLE, A. 1966. The palaeomagnetism of the Sudbury diabase dyke swarm. Can. J. Earth Sci., **4**, pp. 323–332.

LENG, J. 1955. Thesis, University of London.

LOWDON, G. A., STOCKWELL, C. H., TIPPER, H. W., and WANLESS, R. K. 1963. Age determinations and geological studies. Geol. Surv. Can., Paper 62-17.

PARK, J. K. and IRVING, E. 1970. The Mid-Atlantic Ridge near 45°N. XII. Coercivity, secondary magnetization, polarity, and thermal stability of dredge samples. Can. J. Earth Sci., **7**, pp. 1449–1514.

MCGLYNN, J. C. 1970. Geology and economic minerals of Canada (R. J. W. Douglas *Editor*). Geol. Surv. Can., Econ. Geol., Rep. no. 1, 87 p.

ROBERTSON, W. R. 1969. Magnetization directions in the Muskox Intrusion and associated dykes and lavas. Bull. Geol. Surv. Can., **167**.

ROY, J. L., ROBERTSON, W. A., and KEEPING, C. 1969. Magnetic "field free" spaces for palaeomagnetism, rock magnetism and other studies. Can. J. Earth Sci., **6**, pp. 1312–1316.

ROY, J. L., SANDERS, E., et REYNOLDS, J. 1971. Un four électrique pour l'étude des propriétés magnétique des roches. Pub. Dir. Phys. Globe, **42**, no. 8.

SOPHER, S. K. 1963. Palaeomagnetic study of the Sudbury irruptive. Bull. Geol. Surv. Can., **90**, 34 p.

STOCKWELL, C. H. 1936. Great Slave Lake, District of Mackenzie, Northwest Territories. Geol. Surv. Can., Map 378A.

——— 1964. Fourth report on structural provinces, orogenies and time classification of rocks of the Canadian Precambrian Shield. Geol. Surv. Can., Paper 64-17.

STOTT, P. M. 1963. The magnetization of the Red Hill dyke, Tasmania. J. Geophys. Res., **68**, pp. 2289–2297.

STRANGWAY, D. W. 1961. Magnetic properties of diabase dikes. J. Geophys. Res., **66**, pp. 3021–3032.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and DELABIO, R. N. 1970. Age determinations and geological studies. Geol. Surv. Can., Paper 69–2A.